

DEVIN PELLICONE, EIT

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EDUCATION

M.S., Mechanical Engineering, Villanova University, Expected August 2011

B.S. Mechanical Engineering, Temple University, May 2009

EMPLOYMENT

6/10-present Research Assistant, Laboratory for Advanced Thermal and Fluid Systems,
Villanova University

Numerical and Experimental Investigation of Two-Phase Flow and Heat Transfer in Microchannels, Advisor: Dr. Alfonso Ortega

- Created a conjugate, two-phase microchannel heat exchanger model by coupling a created two-phase flow finite difference solver and commercial finite element solver
- Used conjugate model to design and optimize two-phase microchannel heat exchangers for high heat flux dissipation
- Designed experimental flow loop to test commercial and optimized microchannel heat exchangers

9/09-6/10 Research Assistant, Dr. Gerard Jones, Villanova University

Numerical Study of Lightweight Blast Protection Device, Advisor: Dr. Gerard Jones

- Examined the use of viscous liquid-loaded tubes to dissipate the energy generated from a roadside blast
- Developed mathematical models to simulate compressible flow in a small tube
- Developed a CFD model to simulate experimental tests of rapid crushing of a viscous liquid-loaded tube

5/07-9/07 General Accounting Intern, The Hibbert Group, Trenton, New Jersey

- Responsible for general Accounts Payable functions including the paying of invoices and sales tax refunds
- Created five detailed instructional/training manuals on the daily functions performed so that future employees could easily adapt to the position
- Worked directly with the Vice President of Accounting on several cost-cutting projects

TECHNICAL PUBLICATIONS

Pellicone, D., Ortega, A., Delvalle, M., Schon, S., “Simulation of Two-Phase Flow and Heat Transfer in Mini- and Micro-Channels for Concentrating Photovoltaics Cooling,” *Proceedings of the 5th ASME International Conference on Energy and Sustainability (2011)*. (publication pending)

Pellicone, D., Ortega, A., Delvalle, M., “Simulation of Two-Phase Flow and Heat Transfer for Practical Design of Mini- and Micro-Channel Heat Exchangers,” *Proceedings of the 2011 ASME International Mechanical Engineering Congress and Exposition (2011)*. (publication pending)

EXTRACURRICULAR ACTIVITIES

(2008-2009) President (2007-2008) Vice President, Society of Automotive Engineers, Temple University

- Student run organization where an open-wheeled formula race car is designed, fabricated and raced by students
- Responsible for the design of a four-wheel independent suspension including: bell crank design, a-frame geometry, anti-roll bars, wheel load analysis, spring rate analysis, and component packaging
- Oversaw chassis, steering system, braking system, power train, and drive train designs. Implemented small design changes to ensure systems compatibility

(2006 – 2008) Student Tutor, College of Engineering, Temple University

- Tutored students in Engineering Statics, Dynamics, Thermodynamics, Engineering Graphics, and Mechanics of Solids

SKILLS

Fabrication Skills: Mill, Lathe, Drill Press, MIG Welding, TIG Welding

Computer Skills: Proficient in: AutoCAD, SolidWorks, Fluent, Mathcad, Matlab, COMSOL, LabVIEW, Refprop, TecPlot, Microsoft Word, Excel, PowerPoint.

PROFESSIONAL LICENSURE

Engineer in Training, License # ET013788 – Issued in Pennsylvania on 6/18/2010